*BY ORDER OF THE COMMANDER*
*KIRTLAND AIR FORCE BASE*

*AIR FORCE INSTRUCTION 21-101*

*AIR FORCE GLOBAL STRIKE COMMAND*
*Supplement*

*KIRTLAND AIR FORCE BASE*
*Supplement*

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*Maintenance*

# *AIRCRAFT AND EQUIPMENT MAINTENANCE MANAGEMENT*

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This publication supplements Air Force Instruction (AFI) 21-101, *Aircraft and Equipment Maintenance Management*, and AFI 21-101_AFGSCSUP, *Aircraft and Equipment Maintenance*, and is supplemented as follows. It provides additional guidance for implementing and maintaining all logistical responsibilities described in AFI 21-101. It applies to the 377th Air Base Wing (377 ABW) and associate units that perform maintenance on aircraft or munitions and/or operate on or near the flightline. Units assigned to the 58th Special Operations Wing (58 SOW), Air Force Research Lab, Air Force Reserve Command, and Air National Guard are exempt from this supplement. Refer recommended changes and questions about this publication to the Office of Primary Responsibility (OPR) using Air Force (AF) Form 847, *Recommendation for Change of Publication*. Route AF Forms 847s from the field through the appropriate functional's chain of command. Ensure that all records created as a result of processes prescribed in this publication are maintained in accordance with (IAW) AFI33-322, *Records Management and Information Governance Program*, and disposed of IAW Air Force Records Information Management System (AFRIMS) Records Disposition Schedule (RDS). This publication may not be supplemented or

further implemented/extended. Requests for waivers must be coordinated through the OPR of this supplement.

***SUMMARY OF CHANGES***

This document has been substantially revised and must be completely reviewed in its entirety. This revision aligns this supplement to the Air Force and Major Command Guidance Memorandums and Supplements. This revision also supports the 2017 Air Force Directive Publications Reduction Initiative.

**2.4. Maintenance Group Commander (MXG/CC) Responsibilities.** In addition to the responsibilities listed below, the MXG/CC or equivalent must ensure compliance with the maintenance requirements and programs in **Chapter 11** of this instruction. (T-1). Approved variations in the MXG organization does not relieve the MXG/CC of compliance with all the requirements of this AFI. (T-1). The 377 MXG is responsible for the following areas: Munitions, Precision Measurement Equipment Laboratory (PMEL), Airfield Operations, Weather, and Transient Aircraft Alert Services (TAAS). The MXG/CC (or equivalent) will:

2.4.2.1. **(Added)** The 377 MXG Quality Assurance (QA) Superintendent (SUPT) will be the group Environmental, Safety, and Occupational Health (ESOH) focal point and will be the MXG liaison to 377 ABW Safety Office.

2.4.5.1. **(Added)** Ensure severe weather procedures are followed IAW KIRTLANDAFBI 15-101, *Weather Support*.

2.4.9. **(Added)** The 898th Munitions Squadron (898 MUNS) Unit Training Manager (UTM) will perform duties as the MXG Training Manager. The Training Manager will ensure all squadron supervisors are meeting their responsibilities regarding upgrade training of enlisted personnel.

2.4.11.1. **(Added)** For transient aircraft related issues contact the Functional Director at DSN: 246-7934 or the QA Evaluator at DSN: 246-6898.

2.4.43.1. **(Added)** Impoundment procedures for transient aircraft handled by the 377th Maintenance Squadron (377 MXS) TAAS will be outlined in **Chapter 7** of this supplement.

2.4.48.1. **(Added)** 377 ABW does not have assigned Ground Instructional Training Aircraft (GITA) or Training Aircraft Aids (TAA).

2.4.52.1. **(Added)** Assigned units will contact the 377 MXG Technical Order Distribution Office (TODO) to request additional eTools in addition to eTools identified in the 377 MXG Maintenance Standardization and Evaluation Plan (MSEP).

2.4.57.1. **(Added)** All 377 MXG newsletters, crosstalk information and local/HHQ policy announcements can be located at the following 377 MXG SharePoint site: **https://usaf.dps.mil/sites/11775/377%20MXG%20Read%20Files/Forms/AllItems.aspx**

2.4.60.1. **(Added)** 377 MXG weekly conducts "MXG Standup" in conjunction with 377 ABW/CC. Meeting topics include, munitions production, maintenance schedule shortfalls and deviations. Minimum attendees include MUNS supervision, Operations Officer/Maintenance Superintendent and Quality Assurance.

2.4.64.1. **(Added)** To the maximum extent possible, personnel selected for QA positions will have a minimum of 6 months Time on Station (TOS). There are no Mission Design Series (MDS) experience requirements. Exception: No TOS requirement for personnel selected as augmentees. (T-3).

2.5.1.14. **(Added)** PMEL production.

2.5.1.15. **(Added)** Munitions Flight production and delivery activities.

2.5.1.16. **(Added)** Airfield status and impact to 377 ABW and 58 SOW activities.

2.5.1.17. **(Added)** Transient aircraft activities, current and upcoming.

2.5.1.18. **(Added)** Weather and related impacts to base mission.

2.5.1.19. **(Added)** 898 MUNS-related production data, logistics missions, and readiness status.

**4.1.** The 377 MXS is a non-standard organization consisting of the following flights: Munitions, PMEL, and Airfield Operations. The Airfield Operations Flight is organized into the Airfield Management (AM), Weather and TAAS Sections, with the TAAS mission being executed by contract. The Munitions Flight will have its own Munitions Accountable Systems Officer (MASO). Additionally, the 377 MXS does not perform aircraft maintenance. Aircraft specific responsibilities listed in AFI 21-101 are not applicable. AM and Weather responsibilities, which are normally performed by an Operations Support Squadron, are performed by 377 MXS.

4.10.1. **(Added)** Coordinate with the 58 SOW, Plans and Scheduling Section, for flying/maintenance schedule requirements.

**4.14. (Added) Airfield Operations Flight.** Albuquerque International Sunport (KABQ)/Kirtland Air Force Base (KAFB) is a shared use airport. KABQ is responsible for runways and all taxiways except Taxiway H (Hotel). The Federal Aviation Administration (FAA) controls the airspace and movement on runways, taxiways, and helipads.

4.14.1. **(Added)** Airfield Management Operations (AMOPS) is the focal point for all operations on the following military areas: ramps A through E; pads 2 through 5; and taxiway H. Agencies requesting use of parking aprons or pads for exercises, training, painting, fuel cell operations, etc., will coordinate with AMOPS at 846-8335/6 at least 48 hours in advance to allow Notices to Airman (NOTAMs) processing, de-confliction and coordination with numerous base agencies. For short-notice requirements, agencies will contact AMOPS to request ramp space use as soon as possible.

4.14.2. **(Added)** The 377 MXS, Weather Flight, provides weather services to the 377 ABW, 58 SOW, and other units assigned or deployed to KAFB, New Mexico. All supported units shall coordinate with the Weather Flight to either change or request support. When necessary during Protection Level 1 activities outside during non-duty hours, 377 MXS will schedule/provide a Point of Contact (POC) at the weather flight to liaise with 898 MUNS Munitions Control. KIRTLANDAFBI 15-101 further describes the capabilities and utilization of the Weather Flight.

4.14.3. **(Added)** TAAS is contracted and is responsible for providing arrival, processing, and departure services for transient aircraft. They are also responsible for recovery operations of transient aircraft that have a disabled wheel utilizing disabled wheel dollies.

6.3.23. **(Added)** Prior to designation of Chief Inspector and Evaluators, conduct an interview to ensure personnel have proper experience and professionalism.

6.4.7. Review all discrepancies for trends quarterly.

6.4.13. 377 MXG QA SUPT or Chief Inspector will conduct all Evaluator Proficiency Evaluations (EPEs).

6.7.2.10. **(Added)** 377 MXG QA SUPT will publish and distribute the quarterly MSEP plan approved by the 377 MXG/CC, which will also include the Routine Inspection List (RIL), Key Task Listing (KTL), and Acceptable Quality Level (AQL) for all tasks evaluated. 377 MXG MSEP plan is published at the following SharePoint location: **https://usaf.dps.mil/sites/11775/mxgqa/MSEP/Forms/AllItems.aspx?viewpath=%2Fsites%2F11775%2Fmxgqa%2FMSEP%2FForms%2FAllItems.aspx**

6.7.3.1.1. **(Added)** Include the E&I plan in the quarterly 377 MXG MSEP.

6.7.3.3.1.1. **(Added)** Consider historical Personnel Evaluations (PEs), Quality Verification Inspection (QVIs), and other inspections data (especially any items having a pass rate of less than 80 percent); trends, suspected training deficiencies, Tech Data Violation (TDVs), and Detected Safety Violation (DSVs).

6.7.3.3.3.1. **(Added)** Identify the assessment type and minimum number of PEs, QVIs, and Special Inspection (SI) assessments to be conducted monthly. Any evaluations not conducted during the month scheduled will be carried forward to the next month.

6.7.12.1. 377 MXG MSEP Meetings will be conducted IAW AFMAN 21-200, *Munitions and Missile Maintenance Management,* due to exceeding the requirements in this AFI.

6.9.4.3.2. **(Added)** QA will monitor and document initial compliance on Time Compliance Technical Order (TCTOs) and determine depth and frequency of inspection coverage. Coverage is directly related to the complexity of the TCTO as well as to how critical the system or the component is to be modified.

6.9.4.3.3. **(Added)** 377 MXG QA will support the verification, validation and TCTO kit proofing, One-Time Inspections (OTIs) and command-directed modifications. QA, along with the TCTO monitor, will attend all TCTO planning and reconciliation meetings.

6.11.1. **(Added)** OTIs may be equipment condition or procedural compliance oriented and may be continued over a period of time until problems are resolved. QA will support units performing OTIs when requested.

7.1.1. **(Added)** In the event that support equipment or munitions owned by the 377 ABW are involved in an incident where impoundment is mandatory in accordance with this instruction or in an incident where impoundment may be warranted or considered, an Impoundment Authority will be assigned and the 377 MXG QA SUPT will be contacted in order to run the local impoundment checklist where necessary. Should support equipment or munitions being utilized or in the possession of a tenant or Temporary Duty (TDY) unit be involved in an incident that requires impoundment, that unit will take the lead on impoundment processes. The 377 MXG QA SUPT will still be notified in these incidences for coordination and support.

7.4.4. **(Added)** As the 377 ABW does not possess aircraft, the 377 MXG/CC's impoundment responsibilities pertaining to aircraft will deal with transient military aircraft. In the event of such an aircraft, while being handled by 377 MXS Airfield Operations Flight's TAAS, experiences a condition or unknown malfunction making it potentially unsafe for flight, the 377 MXG/CC or representative will immediately coordinate with the owning aircraft's MXG/CC to discuss the need for impoundment and 377 ABW support. If the aircraft is impounded, the owning unit will designate the Impoundment Official and Releasing Authority. 377 MXG/CC will coordinate the safety and security of transient aircraft in need of impoundment with the owning unit's impoundment authority and in accordance with this supplement. Once the owning unit arrives on scene the 377 MXG/CC will continue to ensure appropriate support is coordinated to facilitate the impoundment process.

7.4.5. **(Added)** 377 MXS Airfield Operations will be notified when an impoundment decision has been made regarding transient aircraft.

7.6.1.1. **(Added)** Transient aircraft impoundment will be under the authority of the aircraft's owning unit.

8.1.1. **(Added)** The 377 MXG QA SUPT is the 377 ABW Tool Control Manager (TCM).

8.1.1.1. **(Added)** The 377th Logistics Readiness Squadron Vehicle Management Flight (377 LRS/LGRV) will control and account for their tools IAW AFI 24-302, *Vehicle Management*. The Vehicle Fleet Manager or Vehicle Management Superintendent will develop an Operating Instruction (OI) to specify tool accountability procedures to fit the local mission and physical shop layout. These procedures will be coordinated with the publications office and the 377 ABW TCM for review and approval. After approval, the 377 ABW TCM will approve any changes to the procedure prior to implementation.

8.1.1.2. **(Added)** The 377th Mission Support Group Civil Engineer Division (MSG/CE) and contractors performing work under CE supervision in aircraft maintenance areas will develop a written tool control procedure for their operations. These procedures will be coordinated with the publications office and the 377 ABW TCM for review and approval. Contractor will make any necessary changes required for approval. After approval, the 377 ABW TCM will approve any changes to the procedure prior to implementation by the contractor.

8.1.1.3. **(Added)** Kirtland Aero Club will control and account for their tools IAW **Chapter 7** of AFMAN 34-152, *Air Force Aero Club Operations.*

8.2.1.1.1. **(Added)** Procedures for Two or More Work Centers Controlling Tool Room.

8.2.1.1.1.1. **(Added)** Designate one person from either work center as the supervisor. The supervisor will assume overall responsibility for proper management of all tools, equipment and Composite Took Kit (CTKs) assigned to the tool room, remain accountable for the proper management of tools and equipment, and visit the decentralized location at least quarterly to inventory and inspect the tools and equipment and compare the master documentation maintained in the support section to the tools and equipment.

8.2.1.1.1.2. **(Added)** Long-Term Toolbox Storage. Special function tool kits such as combat distribution team, crash recovery, or other mobility taskings may be stored long term. Tool kits will be inspected using the supervisor inspection criteria and sealed before being stored in an enclosed, controlled, secured area. While in storage, these kits will be inspected every 18 months for inventory content and corrosion prevention. Corrosion preventive compounds may be applied and reapplied per TO 1-1-691, *Cleaning and Corrosion Prevention and Control, Aerospace and Non-Aerospace Equipment,* to prevent corrosion. Tool kits authorized for long-term storage and storage handling will be locally determined.

8.2.1.1.2. **(Added)** Procedures for Control of Test, Measurement, and Diagnostic Equipment (TMDE) Issued/Dispatched in Work Areas.

8.2.1.1.2.1. **(Added)** Control of TMDE will be accounted for in the same manner as all other issued tools and inspected accordingly.

8.2.1.1.3. **(Added)** Procedures for Control of Tooling Included in TCTO/Mod Kits. Tools included with TCTOs/Mod Kits will be accounted for and secured in the same manner as all other issued tools, with the exception of markings, until they are no longer needed and sent back to the issuing organization.

8.2.1.1.3.1. **(Added)** Establish a Master Inventory List (MIL) to conduct inventories and maintain accountability upon receipt of the TCTO/Mod kit. The MIL will include the name of the item, quantity of item(s), and signature of the CTK custodian. Maintain two copies IAW this supplement as long as the kit remains with the unit.

8.2.1.1.4. **(Added)** Procedures for Loaned Tools to Include Issue, Tracking, Duration, and Verification.

8.2.1.1.4.1. **(Added)** Tools loaned to personnel outside of the unit will be tracked using an AF Form 1297, *Temporary Issue Receipt*. Ensure name, duty phone, and organization of the individual signing for the tool/equipment are legible. Additionally, the end of the loan period is documented on the form. At the end of the period the unit custodian will contact the borrowing organization if the tool has not been returned.

8.2.1.1.4.2. **(Added)** List tools/equipment by Equipment Identification (EID) and write "last item" to ensure no more items can be added.

8.2.4.1. **(Added)** Unit Procedures for Replacement Tools. Unit CTK custodians will maintain replacement tools if applicable. These tools must be secured in a lockable container and accounted for on a replacement tool log. The unit CTK custodian will ensure all replacement tools are exchanged one-for-one for broken tools, to include consumable/expendable hand tools. Broken tools will be stored and segregated in a lockable container and accounted for weekly until they can be turned into Defense Logistics Agency (DLA) Disposition.

8.2.5.1.1. **(Added)** The joint inventory will be documented using an Air Force approved system Tool Accountability System or AFGSC Form 140, *CTK Inventory and Control Log*.

8.2.9.4. **(Added)** Rags that are used in Foreign Object Damage (FOD) potential areas or in the performance of any type of aircraft or munitions maintenance must be accounted for. CTK custodians will issue rags in lots of 1, 5, or 10 each.

8.2.9.5. **(Added)** Individuals will sign out rags in the Air Force approved system Tool Accountability System or AFGSC Form 140 and account for each rag upon turn-in. A CTK custodian or tool room individual must ensure the number of rags turned-in is the same as the quantity issued.

8.2.9.6. **(Added)** Dirty rags will be cleaned and/or disposed of as appropriate and according to local hazardous waste procedures. Any questions regarding the storage/disposal of dirty rags shall be directed to the unit Environmental Coordinator or the Base Environmental Management office. As applicable, paper products will be disposed of as appropriate and according to Air Force or local hazardous waste procedures.

8.2.10.1. **(Added)** Tools will be purchased through General Services Administration (GSA) or using the Government Purchase Card (GPC). Unit Commanders will limit the number of personnel authorized to purchase tools to an absolute minimum.

8.3.6.5.2. Tools will be described on the MIL to accurately depict the actual size or type to ensure positive tool control, i.e., screwdriver, common, 6 inches; socket, 3/8" drive X 5/8". Flight/section chiefs or equivalent will update and sign the MIL at least annually or when tools are added, removed, or replaced.

8.3.6.5.3. **(Added)** Empty containers for safety wire, solder, hazardous material, etc. will remain in CTKs until replaced or annotated on the removed tool listing by the CTK custodian.

8.3.11.2.1. **(Added)** Individual Personal Protective Equipment (PPE) items such as hard hats, ear defenders, head sets, reflective belts, etc. are not considered personal tools and are authorized for use, provided they are marked with the owner's first initial, last name, organization and employee number before being used around aircraft, on the flight line or a maintenance area. If these items are part of a CTK, they will be managed in accordance with this supplement.

8.5.2.1.1. **(Added)** If more than one person utilizes the contents of a single CTK, the person initially signing for the CTK is responsible for all contents.

8.5.2.3. Tools and equipment must be visually inventoried before personnel leave the work area.

8.5.2.4. **(Added)** Supervisors are ultimately responsible for tool and equipment accountability and control. When a person removes a tool or piece of equipment, they are responsible for that item until it is returned. Supervisors will be responsible to initiate lost tool procedures.

8.5.2.5. **(Added)** TDY Teams, Department of Energy (DOE), Depot Teams, Factory Representatives, and Contract Field Teams (CFT). Flight chiefs (or equivalent) and CTK custodians will ensure control of tools and equipment used by TDY teams, DOE, depot teams, factory representatives, and CFTs in their area of responsibility. The work center hosting the visiting team will be responsible to brief proper tool control procedures as outlined in this supplement. At a minimum, a method to track tool accountability will be used. Prior to work being accomplished, a full inventory will be completed on any tools or toolboxes that are not CTK compliant. After completion of maintenance and prior to departure of the team and tools, another full inventory will be completed.

8.6.1. 377 ABW units will mark their tools with the nine-digit worldwide identification (WWID) code when possible. If it isn't possible to mark the tool with the nine-digit WWID, the tool must contain the 4-digit WWID and will have identifying character(s) linking the tool to its respective CTK. The first four digits will be IAW **Table 8.1** Future tool marking needs will be addressed to the 377 ABW TCM. Unit CTK Custodian will forward their complete tool-marking scheme to the 377 ABW TCM. Replacement spare tools stored in the tool crib do not need to be etched until placement in a specific CTK.

**Table 8.1. (Added) 377 ABW WWID List (First Four Digits).**

| First Four | Area |
|---|---|
| KRSS | 898 MUNS Munitions Support Section |
| KRSF | 898 MUNS Facility Maintenance |
| KRSE | 898 MUNS VIIDS Section |
| KRSV | 898 MUNS Vehicles Section |
| KRSM | 377 MXS Conventional |
| KRTA | TAAS |
| KRAC | Aero Club |
| KRPL | PMEL |

8.6.1.1.1. **(Added)** Do not de-etch tool part numbers or manufacture numbers.

8.6.1.4.5.2. **(Added)** For vehicle which enter the flightline/airfield, the vehicle key ring will have an identification tag attached, marked with the WWID or vehicle registration number, indicating the total number of pieces. FOD pickers can serve as the ID tag.

8.6.1.4.5.3. **(Added)** To ensure accountability of items (Chocks, FOD Cans, fire extinguishers, etc.), which are assigned to vehicles and enter the flightline/airfield, they will be listed on Page 3 (back-side) of the vehicle's AF Form 1800, *Operator's Inspection Guide and Trouble Report*. This will be used to list items added to the vehicle's inventory requiring the operator's inspection and accountability. For accountability purposes these items will be marked with WWID or vehicle registration number.

8.6.1.4.6. **(Added)** Do not etch, or stamp TMDE in any manner that will affect calibration or the ability to calibrate.

8.6.1.5. 377 MXG/CC does not require use of the EID in addition to AFTO Form 66, *TMDE Bar Codes (Polyester Film)*, for TMDE routinely dispatched from a work center. The use of the AFTO Form 66 alone is sufficient.

8.7.1.1. **(Added)** The originator of the request will be responsible to put together a Locally Manufactured Equipment (LME) request package that contains detailed drawings with measurements, cost estimates & justification for the LME.

8.7.1.2. **(Added)** The LME package will be coordinated through 377 ABW/SE (Safety) and receive the approval of the requesting squadron CC prior to submission to the 377 MXG/MXQ (Quality Assurance) office.

8.7.1.3. **(Added)** If approved, 377 MXG/MXQ (Quality Assurance) will keep a copy and send the original back to the originator to be filed in the continuity book.

8.7.1.4. **(Added)** Unit LME removal requests will follow the same process and justification for removal from current LME listing as required to be added to the unit LME listing.

8.8.1.1.1. **(Added)** CTKs will be kept locked in a secure room when not in use. CTK custodians will control access to tool rooms and be responsible for the Issue/Turn-In of CTK/TAS equipment.

8.8.1.1.2. **(Added)** Issued CTKs will be secured (locked and placed in a secure area free from vehicle or human traffic) when not being utilized.

8.8.1.1.3. **(Added)** Units are authorized to have stationary tool kits, provided they are managed IAW this supplement. Stationary tool kits will have their keys secured in a tool room and issued with the same procedures utilized to issue the normal mobile tool kits.

**8.9. Lost Item/Tool Procedures.** All 377 ABW military, civilian, contractor, subcontractor personnel and associate units that maintain aerospace equipment will comply with the lost tool procedures contained in this supplement.

8.9.2.1. In the event an item/tool is discovered missing after an aircraft has taxied or taken off, and the item/tool was used on the aircraft, AMOPS shall be notified immediately. AMOPS shall contact the aircraft and have it return immediately to parking.

8.9.2.2.1. **(Added)** If an item is lost on or near an aircraft and is not found after the initial search, make notifications to include AMOPS 846-8335, 58th Special Operations Wing Operations Center (WOC) 846-0160 (if incident occurs in 58 SOW areas), and 377 ABW Command Post, 846-3777. If aircraft commander or aircraft crew chief is not present, TAAS personnel will ensure a red "X" is placed in the aircraft or equipment forms of all affected aircraft by qualified personnel. If the item is not found after an exhaustive search including Non-Destructive Inspection using x-ray and/or borescope, the 377 MXG/CC (377 MXG/CD if CC is unavailable) will coordinate with the final authority for releasing the aircraft for flight to clear the red "X" discrepancy in the forms. For Non-Destructive Inspection (NDI) needing x-ray contact 58 MXS Production Superintendent (Production) at 853-7620 for coordination, for borescope contact 58 MXS/MXMP (Propulsion Flight) at 853-2828. In every case, the 377 ABW Command Post will be notified whenever this action is required.

8.9.2.2.2. **(Added)** If an item is lost on a piece of support equipment or another "off-equipment" i.e. item/tool is lost on a piece of support equipment that is in a shop or not on the flight line, the equipment will not be operated until the item is found or the work center supervisor determines the equipment is safe to operate.

8.9.2.3.2.1. **(Added)** The search will continue until found or the search is terminated by the Operations Officer/MX SUPT. Unit Control Centers (UCC) will notify Wing FOD/Dropped Object Prevention (DOP) Manager.

8.9.2.3.2.2. **(Added)** If not found, the UCC will notify the 377 MXG/CC of the missing item/tool (for 377 MXG units).

8.9.2.4.1.1. **(Added)** For Non-Destructive Inspection needing x-ray contact 58 MXS/MXM at 853-7620 for coordination, for borescope contact 58 MXS/MXMP at 846-2828.

8.9.2.5.1.1.1. **(Added)** For non-United States Air Force (USAF) aircraft the owning command/unit shall be notified through the aircraft commander for additional guidance.

8.9.2.6.2.1. **(Added)** All AFGSC Forms 145, *Lost Tool/Object Report,* will be sent to 377 ABW TCM as soon as possible after the search has ended.

8.9.2.6.2.2. **(Added)** If a tool is found on KAFB, contact the 377 ABW TCM.

11.8.3.6.1.1. **(Added)** A subdued non-metallic cord must have a breakaway feature. The use of metal on these items shall be kept at a minimum, (i.e., clip or spiral key ring) and if used; ensure that it cannot be separated from the cord.

11.8.3.6.6. **(Added)** Headgear may be worn within the flightline area for environmental protection; remove and secure headgear when within the area of safety (25' around the aircraft) of an aircraft operating engine(s) and/or Auxiliary Power Units (APU). During windy conditions headgear shall not be worn when engines are operating, even when outside the area of safety.

11.8.3.6.6.1. **(Added)** The boonie hat is considered PPE and under no circumstances will be worn outside Pad 5, the Pad 5 Taxiway, Parking Spot D-7, or the MSA. On the flight line, the chin cord will hold the hat securely in place at all times. Boonie hats will not be rolled, tucked or otherwise formed. Hats will be worn in their natural form. Boonie hats will not be hung around the neck when not being worn.

11.8.3.10.1. No later than one hour prior to an aircraft mission, 898 MUNS personnel will perform a FOD walk of the pad/ramp in use.

11.8.3.11.3. **(Added)** When transient aircraft incur FOD at KAFB, 377 MXS FOD/Dropped Object Program (DOP) Monitors will conduct the investigation and notify the owning organization immediately.

11.8.3.11.3.1. **(Added)** The owning organization is responsible for FOD incidents and investigations on transient aircraft/engines when one of the following conditions applies:

11.8.3.11.3.1.1. **(Added)** FOD is discovered upon arrival at a transient base with no intermediate stops or prior to any engine run.

11.8.3.11.3.1.2. **(Added)** When the owning organization's maintainers are deployed with the aircraft and the FOD is a direct result of transient unit negligence.

11.8.3.15.1. **(Added)** Vehicles normally used for flightline operations should perform periodic inspection throughout the day to ensure tires, bed and undercarriage are clear of foreign objects (FO).

11.8.3.15.2. **(Added)** Emergency vehicles responding to real world emergencies and alert aircrew vehicles responding to Klaxon Calls are not required to perform a FO debris check, but when the emergency/Klaxon Call terminates a FO debris check shall be accomplished prior to proceeding.

11.8.3.15.3. **(Added)** Vehicles driven off paved surfaces shall have the undercarriage clear of debris prior to entering the flightline area; if the vehicle is muddy, the undercarriage shall be washed clean prior to entering the flightline area.

11.8.3.18. With the exception of the FOD picker, items assigned to vehicles will be marked IAW paragraph **8.6.1.4.5.2** and **8.6.1.4.5.3** FOD pickers will be marked at a minimum with vehicle regulation number.

11.8.3.23. **(Added)** Personal electronic or communication devices (e.g., cell phones, portable music/video players, and electronic games) are not authorized for use within the flightline area, munitions areas, and/or other industrial work areas.

11.8.3.23.1. **(Added)** Government communication devices issued for performing official duties must be appropriately marked/identified and can be used within the flightline area. ***NOTE:*** Devices shall be secured or stowed when working within the area of safety of an aircraft operating engines.

11.8.3.24. **(Added)** Umbrellas are not authorized within the flightline area.

11.8.3.25. **(Added)** Footwear with metal cleats and/or taps shall not be worn within the flightline area.

11.8.3.26. **(Added)** Pens, pencils, etc. must be stowed prior to entering the area of safety of an aircraft operating engine(s) and/or APU and stowed prior to engine start when within the area of safety of an aircraft.

11.8.4.2.7. Ensure all maintenance, operations, base support, and contractor personnel who work in, around, or drive through maintenance and/or operational areas are trained on FOD prevention initially and then annually thereafter. Ensure that this training is documented.

11.8.4.2.8. **(Added)** Approves changes and recommendations to the ABW FOD/DOP.

11.8.4.2.9. **(Added)** Appoint the primary and alternate ABW FOD/DOP Monitor.

11.8.4.2.10. **(Added)** Manage investigations for ABW FOD/dropped object mishaps/incidents.

11.8.4.2.11. **(Added)** Ensure mission partner units appoint a unit FOD monitor for their units and are actively involved in the wing's FOD prevention program/committee.

11.8.4.2.12. **(Added)** A FOD/DOP bulletin board will be kept at each unit location. One centrally located board may cover all shops located in a single building. Placement is at the discretion of the individual squadron, but the location must provide the greatest visual access to personnel. The squadron FOD Monitors are responsible for obtaining and maintaining the bulletin board. (Bulletin boards may be combined with already in use unit bulletin boards.) At a minimum, the board should have a copy of the appointment memorandums of the wing, squadron, and/or flight FOD/DOP Monitors with the proper signature authority, and the ABW FOD prevention poster.

11.8.5.6. Serves as POC for all FOD/dropped object prevention issues within the ABW and acts as liaison between ABW units and the 58 SOW FOD/DOP Program Manager (58 SOW/CV).

11.8.5.7. **(Added)** Enforce weekly FOD inspections. Units are responsible for weekly FOD inspections of their functional areas including aircraft parking ramps, taxi lanes, hangers, areas where equipment is stored, maintained, and/or serviced, and any other area which the presence of FO debris may damage aircraft or equipment; document inspections on a unit locally developed checklist and maintain in unit continuity book. This includes all units that perform maintenance on the flightline, and that operate vehicles or other equipment on the flightline.

11.8.5.8. **(Added)** Coordinates with the 58 SOW FOD/DOP Program Manager (58 SOW/CV) and provides ABW attendees with copies of the meeting minutes.

11.8.5.9. **(Added)** Analyzes 377 ABW FOD/DOP trend data and lost tool reports.

11.8.5.10. **(Added)** Coordinates with 377 ABW unit FOD/DOP Monitors to ensure publicity and awareness materials are disseminated and being utilized.

11.8.5.11. **(Added)** Review and analyze all unit FOD mishap reports and other data for trends that identify areas requiring management action.

11.8.6.15. **(Added)** The 58 SOW/MXQ has primary responsibility for investigating FOD/DOP mishaps/incidents involving 58 SOW aircraft. 377 MXS FOD/DO Monitor will investigate all transient aircraft mishaps.

11.8.6.16. **(Added)** Mission partners shall investigate and report FOD/DOP mishaps/incidents IAW their applicable command instructions.

11.8.6.17. **(Added)** The 377 ABW FOD/DOP monitor will submit all FOD/DOP incidents with the exception of minor sand nicks and scratches to AFGSC FOD/DOP Monitors.

11.8.6.18. **(Added)** The 377 ABW FOD/DOP monitor will ensure the investigation has been completed and all data for the FOD/DOP report is accurate and complete before closing the report.

11.8.6.19. **(Added)** If the FOD incident is deemed a mishap IAW DAFI 91-204, *Safety Investigations and Reports*, the Wing FOD monitor will still initially report the FOD incident then work in tandem with the Wing Safety Office to properly report the mishap.

11.8.7.2.13.1. **(Added)** Wing FOD Awards Program. The purpose of the Wing FOD Awards program is to promote FOD awareness and to enhance the effectiveness of the FOD prevention effort. The wing has the Golden Bolt program.

11.8.7.2.13.1.1. **(Added)** The Golden Bolt Award promotes FOD awareness and attention to detail to identify and eliminate potential sources of FOD. The Golden Bolt is placed in various places including high traffic areas, cargo pads, maintenance shops and vehicles. The Golden Bolt is never placed in an area where it could become a FOD hazard. The 377 ABW FOD/DOP Manager or Quality Assurance personnel will place the object and ensure it is found or removed. The award is a certificate and a 1-day pass from the 377 ABW/CV.

11.8.7.3. **(Added)** When meetings are required, Mission Partner units FOD/DOP Committee Meeting/Combined ABW FOD/DOP Committee Meeting. Mission Partner units will coordinate with the ABW FOD monitor to ensure participation and compliance with ABW program. Mission partner Units will have FOD program managers or alternates represented in the ABW committee meeting. The meeting is held quarterly and conducted by the 58 SOW and the 377 ABW. The 377 ABW/ CV, or 377 MXG/CC when the CV is not available, co-chairs the committee.

11.8.9. **(Added) 377 ABW unit commanders shall:**

11.8.9.1. **(Added)** Implement FOD/dropped object prevention practices and procedures outlined in this supplement.

11.8.9.2. **(Added)** Publicize FOD/DOP events, statistics, incidents, and all other pertinent information in an effort to encourage awareness and the importance of FOD/dropped object prevention.

11.8.9.3. **(Added)** Appoint a primary and alternate unit FOD/DOP Program Monitors, forward a letter of appointment to the 377 ABW FOD/DOP Monitor workflow at **377mxg.qa@us.af.mil**.

11.8.10. **(Added) Mission Partners Unit/Subordinate Units FOD/DOP Program Monitor shall:**

11.8.10.1. **(Added)** Act as focal points for all FOD/dropped object prevention issues within their unit and, as required, communicate issues to the 377 ABW FOD/DOP Monitor.

11.8.10.2. **(Added)** Maintain unit FOD/dropped object prevention bulletin boards in each work center and ensure information is current. One centrally located board may cover all shops in a single building. The board should be located in a manner that provides the greatest visual access to all personnel. At a minimum, the board should have a copy of the appointment memorandums of the wing, squadron, and/or flight FOD/DOP Monitors with the proper signature authority, a FOD prevention poster, and a copy of the latest combined meeting minutes.

11.8.10.3. **(Added)** Maintain a FOD/DOP continuity book, at a minimum; include letter of appointment signed by the squadron/unit commander, appointment letter for the 377 ABW POCs, list of FOD references, combined committee meeting minutes, 377 ABW quarterly/monthly meeting minutes, as required, FOD/DOP cross tell, and weekly inspections.

11.8.10.4. **(Added)** Ensure FOD/DOP initial training and recurring training is conducted and documented.

11.8.10.5. **(Added)** Ensure all contractors and/or subcontractors under their control requesting personnel or vehicle access to the flightline areas comply with all applicable FOD prevention practices.

11.8.11. **(Added) 377 ABW/SE shall:**

11.8.11.1. **(Added)** Assist the 58 SOW/SE with FOD/DOP mishap/incident investigations.

11.8.11.2. **(Added)** Act as liaison between the 58 SOW/SE and 377 ABW/CC.

11.8.11.3. **(Added)** Assist the 377 ABW FOD/DOP monitor (377 MXG/MXQ), as required, with investigation and reporting transient FOD/DOP mishap/incidents.

11.9.1.2.1. **(Added)** The 377 ABW FOD monitor will be designated as the wing DOP monitor.

11.9.1.3. **(Added)** The 377 MXG/CC will ensure all assigned maintenance personnel are briefed on the DOP on a recurring basis, at least annually. Annotate DOP training in appropriate training records or in an automated system.

11.9.1.4. **(Added)** Personnel will ensure all equipment used to upload/offload cargo is accounted for prior to leaving aircraft and/or area.

11.9.2. The 377 ABW DOP monitor will be responsible to investigate dropped objects from a transient aircraft. The 377 ABW DOP monitor will provide the home station DOP monitor with sufficient data to generate a report for trending and tracking purposes.

11.9.3. The 377 ABW DOP monitor will be responsible to investigate dropped objects from a transient aircraft. The 377 ABW DOP monitor will provide the home station DOP monitor with sufficient data to generate a report for trending and tracking purposes.

11.28.2.4.4. **(Added) 377 ABW Responsibilities:**

11.28.2.4.4.1. **(Added)** Provide Crash Damage or Disabled Aircraft Recovery (CDDAR) support to the 58 SOW/WOC IAW the support agreement between the 377 ABW and 58 SOW.

11.28.2.4.4.2. **(Added)** KAFB Command Post (KCP) will coordinate 58 SOW/WOC requests for CDDAR support to include composite damage and hazard abatement.

11.28.2.4.4.3. **(Added)** 377 LRS will provide the following if the 58 SOW has a CDDAR equipment shortfall:

11.28.2.4.4.3.1. **(Added)** Aircraft defuel capabilities.

11.28.2.4.4.3.2. **(Added)** Transportation requirements for personnel, equipment, and material.

11.28.2.4.4.3.3. **(Added)** Tractor, trailer and qualified driver/operator.

11.28.2.4.4.3.4. **(Added)** Vehicle maintenance support.

11.28.2.4.4.3.5. **(Added)** Shoring materials.

11.28.2.4.4.4. **(Added)** 377 MSG/CE will provide large capacity crane (over 7.5 tons) and other necessary heavy equipment including drivers/operators. CE shall procure needed equipment from off-base resources.

11.28.2.4.4.5. **(Added)** The 377th Security Forces Group will provide security of the mishap/accident site.

11.28.2.4.4.6. **(Added)** The Bio Environmental Engineer will brief recovery team members of potential site hazards, identify Personal Protective Equipment requirements, and train members on the proper use of PPE.

11.28.2.4.5. **(Added)** Transient Aircraft/Alert Services will provide disabled aircraft recovery of transient aircraft IAW KAFB Performance Work Statement paragraph 1.6.3. “remove transient aircraft that are disabled on runways, taxiways, aircraft aprons, aircraft parking areas, and aircraft pads utilizing disabled wheel dolly(s).”

11.28.2.4.6. **(Added)** The 377 ABW POC/lead for transient aircraft is the 377 MXS/OO. Requests for support from the 58 SOW will flow from the KCP to the 58 SOW/WOC. For incidents on Albuquerque Sunport property, the 377 MXS/OO will coordinate actions with Sunport Operations.

11.28.2.4.7. **(Added)** If recovery operations of transient aircraft are beyond 377 ABW capabilities, KCP will notify the transient aircraft home station for CDDAR support and provide contact information for the 377 MXS/OO. The 377 MXG/CC will assume Recovery Operations Chief responsibilities.

11.28.2.4.8. **(Added)** 377 ABW will provide all support for the home unit CDDAR team listed in paragraphs **11.28.2.4.4.4 through 11.28.2.4.6** The 377 MXS/OO will be the main POC for the visiting CDDAR team.

JASON F. VATTIONI, Colonel, USAF
Commander

## Attachment 1

## GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

***References***

AFI 21-101, *Aircraft and Equipment Maintenance Management*, 16 January 2020

AFI 21-101_AFGSCSUP, *Aircraft and Equipment Maintenance Management*, 19 January 2021

AFMAN 34-152, *Air Force Aero Club Operations*, 04 June 2019

KIRTLANDAFBI 15-101, *Weather Support*, 9 July 2019

***Adopted Forms***

AF Form 1800, *Operator's Inspection Guide*

***Abbreviations and Acronyms***

**ABW**—Air Base Wing

**AF**—Air Force

**AFGSC**—Air Force Global Strike Command

**AM**—Airfield Management

**AMOPS**—Airfield Management Operations

**CC**—Commander

**DOE**—Department of Energy

**DRMS**—Defense Reutilization and Marketing Service

**GPC**—Government Purchase Card

**GSA**—General Services Administration

**H**—Hotel

**KABQ**—Albuquerque International Sunport

**KAFB**—Kirtland Air Force Base

**KCP**—Kirtland Command Post

**KIRTLANDAFBI**—Kirtland Air Force Base Instruction

**MASO**—Munitions Accountable Systems Officer

**MOO**—Maintenance Operations Officer

**NMC2**—Nuclear Munitions Command and Control

**NOTAM**—Notices to Airman

**SOW**—Special Operations Wing

**TAAS**—Transient Aircraft Alert Services

**TCM**—Tool Control Manager

**TPE**—Trainer Proficiency Evaluation

**TOS**—Time on Station

**UCC**—Unit Control Center

**WOC**—Wing Operations Center